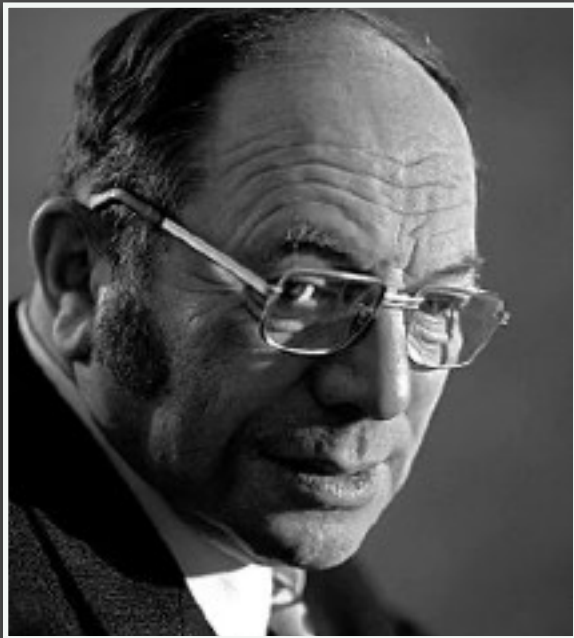






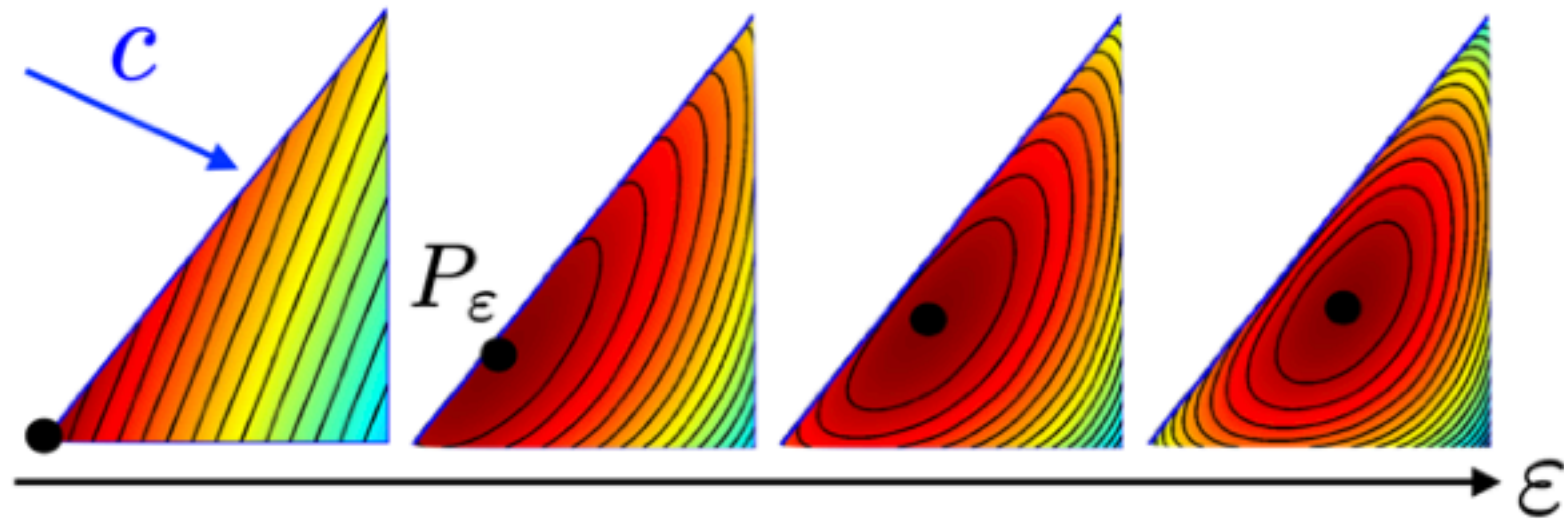


# Kantorovich's Problem:

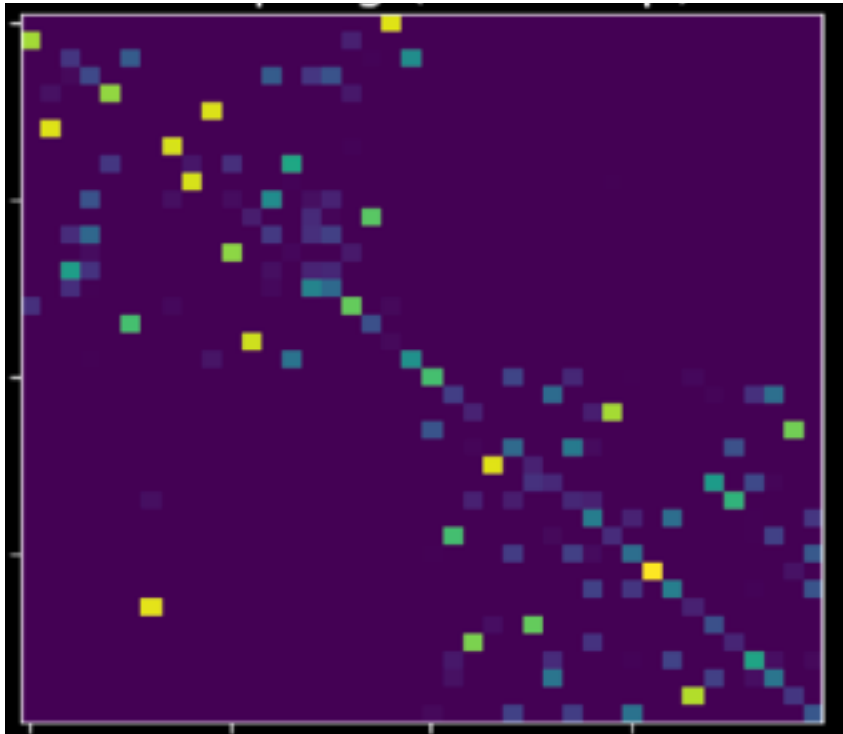
(Discrete version + entropy regularized)

$$\min_{\pi \in \Pi(\mathbf{a}, \mathbf{b})} \langle \pi, \mathbf{C} \rangle - \varepsilon H(\pi)$$

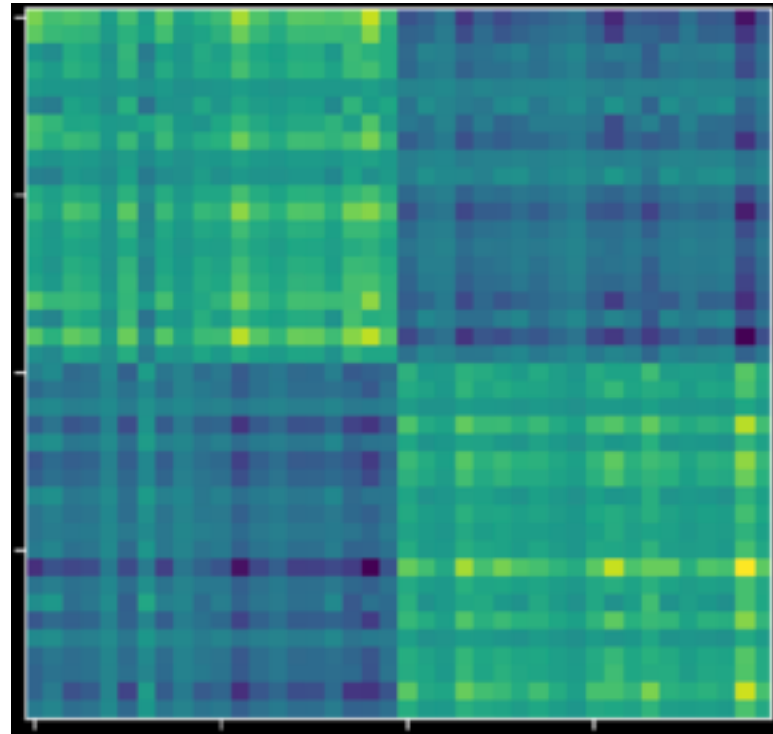




**Figure 4.1:** Impact of  $\varepsilon$  on the optimization of a linear function on the simplex, solving  $\mathbf{P}_\varepsilon = \operatorname{argmin}_{\mathbf{P} \in \Sigma_3} \langle \mathbf{C}, \mathbf{P} \rangle - \varepsilon \mathbf{H}(\mathbf{P})$  for a varying  $\varepsilon$ .

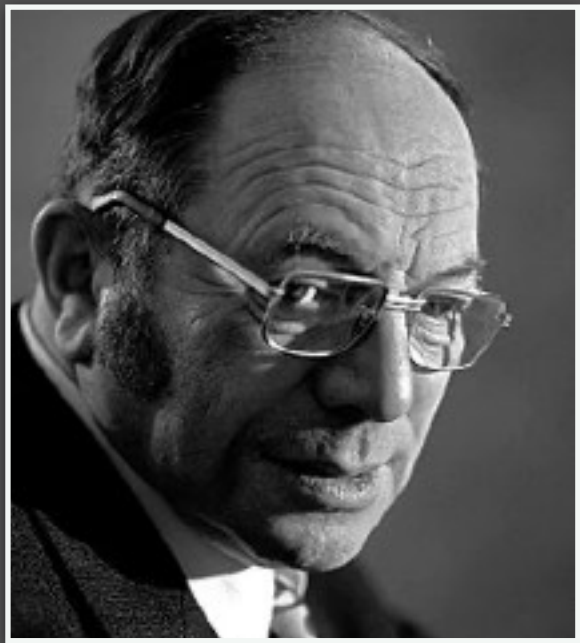


among OT



among all feasible

# Entropic OT Formulation



## Kantorovich's Problem:

(Discrete version + entropy regularized)

$$\min_{\pi \in \Pi(\mathbf{a}, \mathbf{b})} \langle \pi, \mathbf{C} \rangle - \varepsilon H(\pi)$$

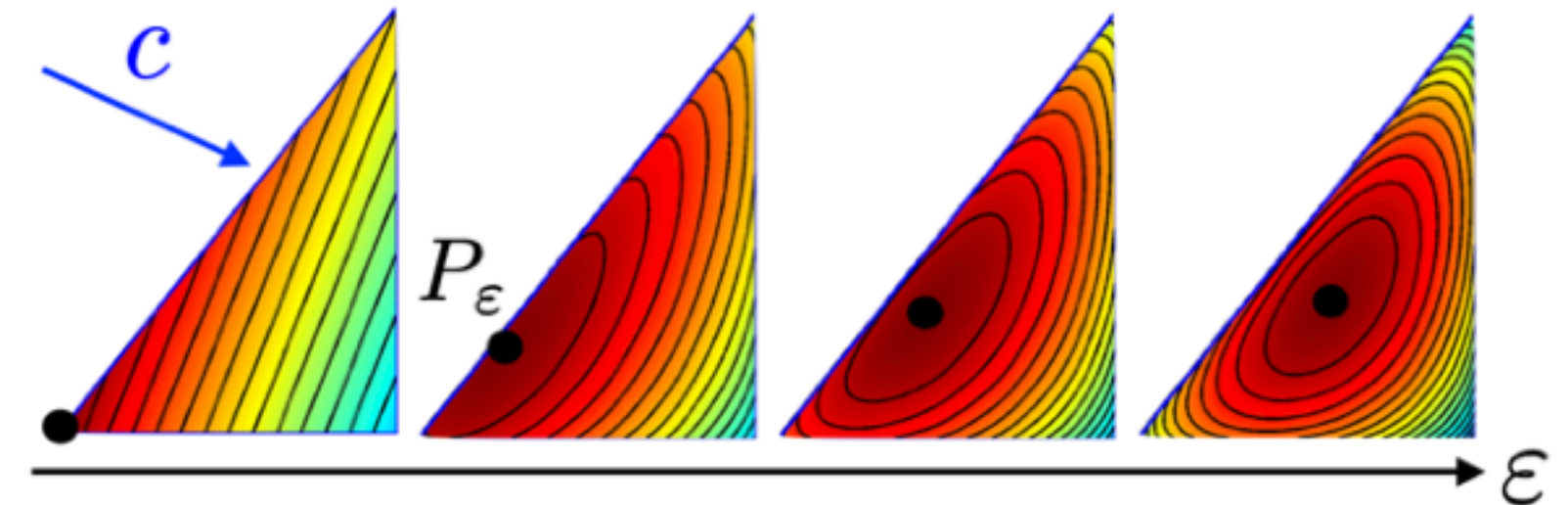
**Theorem** [Convergence of entropic OT]:

The unique solution of the EOT problem converges to max-entropy optimal solution of the original problem:

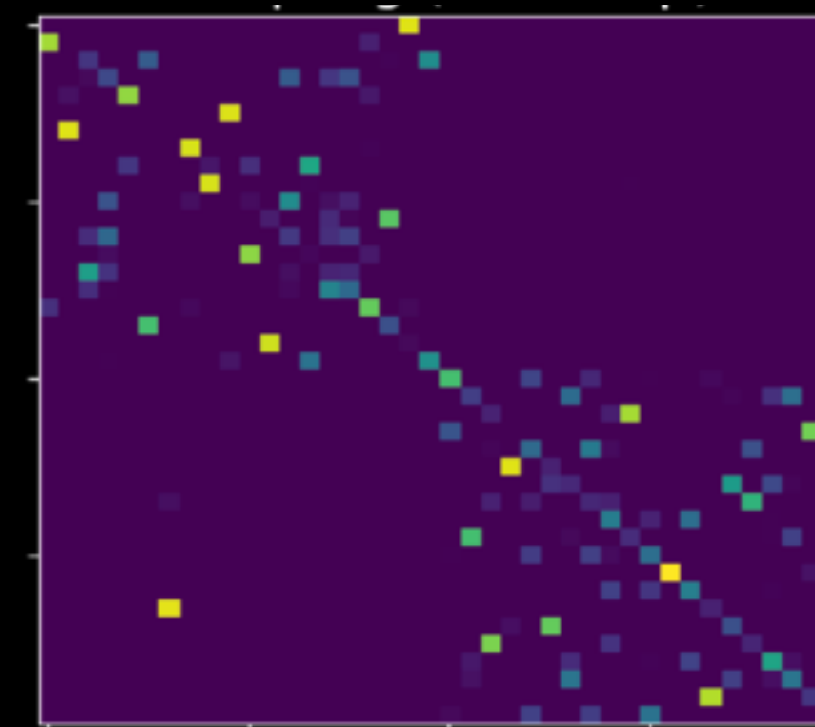
$$\pi_\varepsilon \xrightarrow{\varepsilon \rightarrow 0} \operatorname{argmin}_{\pi} \{ -H(\pi) \mid \pi \in \Pi(\mathbf{a}, \mathbf{b}), \langle \pi, \mathbf{C} \rangle = L_C(\mathbf{a}, \mathbf{b}) \}$$

And in particular,  $\operatorname{OT}_\varepsilon(\alpha, \beta) \xrightarrow{\varepsilon \rightarrow 0} \operatorname{OT}(\alpha, \beta)$ .

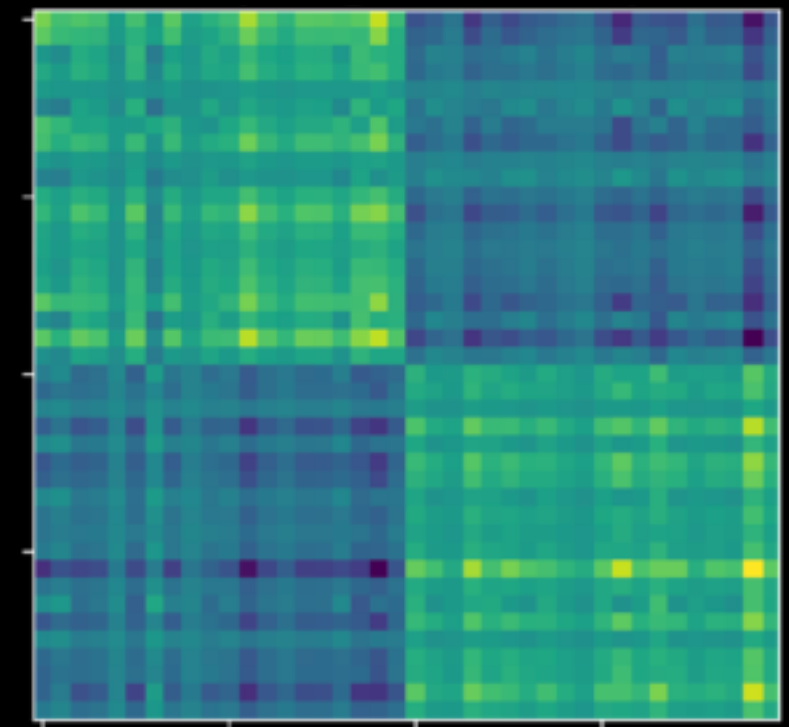
Also,  $\pi_\varepsilon \xrightarrow{\varepsilon \rightarrow \infty} \mathbf{a} \otimes \mathbf{b}$



**Figure 4.1:** Impact of  $\varepsilon$  on the optimization of a linear function on the simplex, solving  $\mathbf{P}_\varepsilon = \operatorname{argmin}_{\mathbf{P} \in \Sigma_3} \langle \mathbf{C}, \mathbf{P} \rangle - \varepsilon H(\mathbf{P})$  for a varying  $\varepsilon$ .



Maximal entropy coupling  
among OT solutions



Maximal entropy coupling  
among all feasible solutions

0 ←  $\varepsilon$  →  $\varepsilon \rightarrow \infty$



EOT solution has a nice form